

Data Structures & Algorithms - University Notes

Part 1

Introduction

A data structure organizes and stores data efficiently.

Types

Primitive: int, float, char, bool. Non-Primitive: Array, Linked List, Stack, Queue, Tree, Graph.

Complexity

Time complexity measures execution time growth. Space complexity measures memory usage.

Big O

$O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$.

Applications

Used in databases, operating systems, networking, AI, and web applications.